



Enterprise · Integrated · Connected

Global Business Operating System Production MVP — Comprehensive Build Plan

Document	EIC Enterprise MVP Build Plan v1.0
Prepared for	Engola Innocent — CEO & Founder
Website	www.eic.global
Date	April 2026
Classification	STRICTLY CONFIDENTIAL
Target Market	African SMEs & Global Enterprises

"One Platform. Every Business. Every Country. Every Currency."

Table of Contents

01	Executive Summary & Vision	3
02	Brand Identity & Design System	4
03	Platform Architecture Overview	5
04	Module 1 — Smart Dashboard	7
05	Module 2 — Accounting & Finance	8
06	Module 3 — EFRIS / Tax Compliance (URA)	9
07	Module 4 — Mobile Money & Pi Wallet → Crypto Wallet	10
08	Module 5 — Banking Integration	11
09	Module 6 — CRM & Sales Pipeline	12
10	Module 7 — IT Asset Management	13
11	Module 8 — Crypto Exchange (BTC, ETH, USDT, SOL, BNB)	14
12	Module 9 — ARIA AI Business Assistant	15
13	Module 10 — HR & Payroll	16
14	Module 11 — Inventory & Supply Chain	17
15	Module 12 — Project Management	18
16	Payment Integration Architecture	19
17	PCI DSS Compliance Framework	21
18	Security Architecture	23
19	Database Schema & API Design	25
20	DevOps, CI/CD & Infrastructure	27
21	Go-to-Market & Pricing Strategy	29
22	Build Phases & Timeline	31
23	Pre-Launch Checklist	33
24	Legal, Privacy & Regulatory	34

1

EXECUTIVE SUMMARY & VISION

The EIC Vision

EIC Enterprise is a cloud-native, AI-powered, all-in-one business operating system engineered to serve every category of business globally — from a one-person SME in Kampala to a multi-national corporation operating across 50 countries. It unifies accounting, tax, payments, banking, HR, CRM, IT management, project management, inventory, and crypto into a single premium SaaS platform.

Founded and led by Engola Innocent in Uganda, EIC is built with the philosophy that world-class enterprise software should not be exclusive to Fortune 500 companies. The platform targets the 550 million African SMEs, the 400 million global micro-enterprises, and the growing class of digital-native businesses who demand mobile-first, AI-native tools.

Core Differentiators

- African-First Architecture:** Built for low-bandwidth environments, supports MTN MoMo, Airtel Money, M-Pesa natively alongside SWIFT/SEPA for global payments.
- Regulatory Intelligence:** Deep EFRIS/URA integration for Uganda tax compliance; extendable to KRA, FIRS, SARS and global tax bodies.
- AI-Native from Day One:** ARIA (AI assistant) is embedded across every module — not bolted on — providing contextual insights, anomaly detection, and auto-reporting.
- Crypto-Enabled Treasury:** Full BTC, ETH, USDT, SOL, BNB wallet and payment acceptance built in; real-time fiat conversion for every supported currency.
- Multi-Currency, Multi-Country:** Every financial entity is currency-aware. 170+ currencies supported. Automatic FX rounding, multi-jurisdiction tax, and IFRS-aligned reporting.
- Zero Hardware Dependency:** Progressive Web App (PWA) that runs on a UGX 200,000 Android phone. Offline-capable for markets with intermittent connectivity.

Business Model Summary

Tier	Target	Price / Month (USD)	Key Modules
Starter	1–5 employees	\$29	Dashboard, Accounting, EFRIS, Mobile Money
Growth	6–50 employees	\$99	All Starter + CRM, Banking, HR, Inventory
Business	51–500 employees	\$299	All Growth + IT Mgmt, Projects, Crypto
Enterprise	500+ / Custom	Custom	All modules + White-label + SLA + Dedicated
Marketplace	Apps / Integrations	Revenue Share	Third-party add-ons

Visual Identity

EIC operates a strictly controlled dark-mode-first brand. The palette communicates authority, technological sophistication, and financial trust. Every UI surface uses glassmorphism cards, subtle depth layering, and a single gold accent — never cluttered, always intentional.

Colour Palette

Token	Hex	Role	Usage
--bg	#06091A	Deep Space Navy	Page / app background
--card	#0D1436	Card Dark	All glassmorphism card backgrounds
--border	#1E2448	Dim Border	Card edges, table lines, dividers
--gold	#F5A623	Electric Gold	Primary accent, CTAs, active nav, headings
--emerald	#00C48C	Emerald	Positive indicators, success, profit
--white	#F7F7F7	Primary Text	Body copy, labels, values
--grey	#A0A4B8	Muted Grey	Secondary text, placeholders
--danger	#E53E3E	Danger Red	Errors, negative P&L, overdue alerts
--info	#3B82F6	Info Blue	Informational chips, tooltips

Typography

Role	Font Family	Weights	Usage
Display / Headings	Plus Jakarta Sans	600, 700, 800	Module titles, hero text, stat values
Body / UI	Inter Tight	400, 500, 600	All body copy, labels, table data
Monospace	JetBrains Mono	400, 500	Code blocks, API keys, account numbers

Logo & Identity System

The EIC wordmark uses a custom weight of Plus Jakarta Sans with a 4pt letter-spacing, rendered in Electric Gold on Deep Space Navy. The logomark consists of three interlocking hexagonal rings representing Enterprise connectivity across three pillars: Finance, Operations, and Intelligence. A linear gradient from Gold to Emerald is applied for motion/digital contexts. The favicon is a single gold hexagon with the letterform 'E' cut out in navy.

Brand Voice & Positioning

Tone: Confident, precise, empowering. Never corporate-jargon. Never casual. Tagline: **'One Platform. Every Business. Every Country. Every Currency.'** Hero video concept: Aerial drone footage over Kampala at golden hour transitions into Lagos skyline, then Nairobi, Dubai, London — overlaid with animated data flows and the EIC gold wordmark fading in. Music: minimalist Afrobeats-electronic fusion. Duration: 90 seconds.

Technical Stack

Layer	Technology	Rationale
Frontend Framework	React 18 + TypeScript 5	Type safety, ecosystem maturity, SSR capability
Build Tool	Vite 5	Sub-second HMR, tree-shaking, ESM native
Styling	Tailwind CSS v4 + CSS Variables	Utility-first, token-driven theming
Routing	React Router v6 / TanStack Router	Code-split lazy routes, nested layouts
State	Zustand + React Query (TanStack)	Minimal boilerplate; server-state caching
Charts	Recharts + D3 (advanced)	Declarative charts with D3 escape hatches
Animation	Framer Motion	Physics-based motion, layout animations
Icons	Lucide React	Consistent 24px stroke icons, tree-shakeable
Backend	Node.js 20 + Express 5	Proven, lightweight, massive ecosystem
ORM	Drizzle ORM	TypeScript-first, zero-overhead SQL generation
Database	PostgreSQL 16 (primary)	ACID, JSONB, full-text, row-level security
Cache	Redis 7	Session store, rate-limit counters, pub/sub
Auth	JWT + Refresh Tokens + OAuth2	Stateless auth with secure rotation
File Storage	AWS S3 / Cloudflare R2	Multi-region, CDN-backed, cost-efficient
Email	Resend (transactional)	React Email templates, SPF/DKIM compliant
SMS / OTP	Africa's Talking + Twilio	Local carrier routing in Africa + global fallback
AI	Anthropic Claude claude-sonnet-4-20250514	MIRA assistant, document AI, anomaly detection
Payments — Africa	Flutterwave + MTN MoMo SDK + Airtel SDK	Local acquirer coverage 35+ African markets
Payments — Global	Stripe (cards) + Pesalink + SWIFT	PCI SAQ-A, global card acceptance
Crypto	Coinbase Commerce + BitPay + Web3.js	BTC/ETH/USDT/SOL/BNB multi-network
DevOps	Docker + Kubernetes (K8s) + GitHub Actions	Container orchestration, zero-downtime deploys
CDN / Edge	Cloudflare Workers + CDN	Global edge, DDoS protection, WAF
Monitoring	Datadog APM + Sentry + PagerDuty	Full observability stack, on-call alerting
Search	MeiliSearch (self-hosted)	Sub-50ms fuzzy search across all entities

High-Level System Architecture

EIC uses a modular monolith backend deployed inside Docker containers orchestrated by Kubernetes. Each business domain (Accounting, CRM, Payments, etc.) is a self-contained module with its own service layer, but all modules share a single PostgreSQL instance with row-level security enforcing tenant isolation. The frontend is a React SPA delivered via Cloudflare CDN with route-based code-splitting keeping initial bundle under 200KB gzipped.

Tier	Components	Scale Strategy
Client	React PWA (Web), React Native (Mobile Q3)	CDN edge delivery, service worker cache
API Gateway	Cloudflare Workers + rate limiter + WAF	1M req/s capacity, geo-routing
App Servers	Node.js + Express 5 (K8s pods)	Horizontal auto-scale on CPU/memory
Business Logic	Domain modules (12 modules)	Isolated service layers, shared DB
Data	PostgreSQL 16 (primary) + Redis 7 (cache)	Read replicas, connection pooling (PgBouncer)
AI Layer	Anthropic API + internal vector store	Async processing, batch embeddings
External APIs	Flutterwave, Stripe, MTN, Airtel, URA EFRIS	Circuit breakers, retry queues
Storage	Cloudflare R2 (files) + S3 (backups)	Geo-redundant, AES-256 at rest
Observability	Datadog + Sentry + structured logs	P99 alerting, distributed tracing

Multi-Tenancy Model

EIC uses a shared-database, schema-per-tenant approach for Starter/Growth tiers and dedicated PostgreSQL schemas for Business/Enterprise tiers. Row-Level Security (RLS) policies on every table enforce that queries only ever return data belonging to the authenticated tenant. The **organisation_id** claim from the JWT is verified at the database connection layer — SQL injection cannot bypass tenant isolation because the `org_id` is set as a Postgres session variable before any query executes.

4

MODULE 1 — SMART DASHBOARD

The command-centre of EIC. Surfaces the most critical business KPIs at a glance using real-time data aggregated from all other modules. Designed to be understood in under 3 seconds — the CEO test.

Key Features

- 4× 3D-tilt StatCards: Revenue, Expenses, Net Profit, Tax Due — with sparkline trend arrows
- Revenue vs Expenses area chart (Recharts, last 12 months, interactive tooltips)
- Tax Breakdown donut chart (VAT, WHT, Corporate Tax, PAYE)
- Monthly Cash Flow waterfall bar chart
- Recent Transactions table: amount, party, type, status badge, timestamp
- Quick Actions panel: New Invoice, Record Payment, File Return, Send Report
- Live EAT/UTC clock in header; notification bell with priority inbox
- ARIA AI insight cards: anomaly alerts, cash-flow forecasts, overdue reminders
- Collapsible sidebar: full-width (240px) ↔ icon-only (60px) with smooth CSS transition
- Multi-currency toggle: display all values in any of 170+ currencies

Data Integration: Accounting module, Banking module, EFRIS module, CRM pipeline value

5

MODULE 2 — ACCOUNTING & FINANCE

A full double-entry accounting engine built to IFRS standards. Handles invoicing, bills, journals, bank reconciliation, multi-currency ledgers, and generates publication-ready financial statements.

Key Features

- Chart of Accounts (CoA): fully customisable, hierarchical, IFRS/local GAAP classes
- Double-entry Journal Engine: every transaction auto-posts debit + credit entries
- Invoicing: create, send, track, collect PDF invoices with branded templates
- Bills & Expenses: supplier bills, receipt scanning via AI (OCR), expense approval workflow
- Bank Reconciliation: automated matching against bank statement imports
- P&L; Statement, Balance Sheet, Cash Flow Statement — exportable to PDF/Excel
- Aged Receivables & Payables: overdue tracking with colour-coded aging buckets
- Multi-currency: FX rates via Open Exchange Rates API; realised/unrealised FX gain/loss auto-journal
- Budget vs Actuals: department-level budget tracking with variance analysis
- Audit Trail: every journal entry is immutable; only adjusting entries allowed
- AI Categorisation: ARIA suggests account codes for new transactions using ML

Data Integration: All payment modules, Banking module, EFRIS (tax journals), HR/Payroll

6

MODULE 3 — EFRIS / TAX COMPLIANCE (URA)

Deep integration with Uganda Revenue Authority's Electronic Fiscal Receipting and Invoicing System (EFRIS). Automates VAT calculation, e-invoice generation, e-receipt issuance, and return filing. Designed to eliminate manual compliance risk.

Key Features

- Automatic e-Invoice generation on every sale (EFRIS API v2.0)
- Real-time e-Receipt issuance with QR code for buyer verification on URA portal
- VAT Rate Engine: 18% Standard, 0% Zero-Rated, Exempt, 6% WHT — auto-applied by item class
- Monthly VAT Return auto-preparation from posted journals
- One-click Return Filing directly to URA EFRIS (POST /submitReturn)
- Compliance Score dashboard: green/amber/red with actionable remediation steps
- EFRIS Device registration management (Device No., TIN validation)
- Submission history table with EFRIS acknowledgement numbers
- Penalty calculator: auto-compute late-filing penalties and interest
- Multi-country Tax Engine (extensible): pluggable adapters for KRA (Kenya), FIRS (Nigeria), SARS (SA)
- WHT Certificate generation for applicable supplier payments

Data Integration: Accounting (sales/purchase journals), Banking (payment confirmation)

7

MODULE 4 — MOBILE MONEY & CRYPTO WALLET

Replaces the legacy Pi Wallet concept with a full-spectrum digital treasury: African mobile money (MTN MoMo, Airtel Money, M-Pesa) natively integrated alongside a multi-chain crypto wallet supporting BTC, ETH, USDT, SOL, and BNB. All balances reconcile to the accounting ledger in real time.

Key Features

- MTN Mobile Money SDK: C2B collections, B2C disbursements, balance query, transaction status
- Airtel Money SDK: same capability set as MTN, auto-fallback routing
- M-Pesa (Safaricom Daraja API): STK Push for collections; B2C for disbursements
- Send Money form: recipient number, network auto-detect, amount, reason, 2-second async confirm
- Crypto wallet: BTC (Bitcoin), ETH (Ethereum), USDT (ERC-20 & TRC-20), SOL (Solana), BNB (BSC)
- Accept crypto payments via Coinbase Commerce API: QR code invoice, webhook confirmation
- Real-time price feed: CoinGecko API (BTC, ETH, SOL, BNB → UGX, KES, NGN, USD)
- Crypto-to-fiat conversion calculator with live spread display
- Transaction history with on-chain explorer links (Etherscan, Solscan, BscScan)
- Centenary Bank settlement account (****8770): funds settle via Flutterwave → Centenary
- Wallet-to-wallet transfers within EIC ecosystem (zero-fee internal routing)
- Anti-Money Laundering (AML) screening: transactions >\$1,000 flagged for KYC review

Data Integration: Accounting (auto-journal on settlement), EFRIS (if taxable transaction)

8

MODULE 5 — BANKING INTEGRATION

Connects EIC to Ugandan commercial banks and, via SWIFT/SEPA adapters, to global correspondent banks. Enables account balance views, statement imports, and fund transfers without leaving the platform.

Key Features

- Bank account cards: Centenary Bank, Stanbic Bank, DFCU Bank — balance + last-sync timestamp
- OFX/CSV bank statement import with AI transaction categorisation
- Fund Transfer form: internal (same bank), local (other Ugandan bank via RTGS/EFT), international (SWIFT)
- Bank Reconciliation: side-by-side ledger vs statement view with auto-match and exception list
- Standing orders and direct debit management
- Cheque management: issue, void, reconcile
- Open Banking API layer (for future PSD2 / Open Finance compliance)
- DFCU/Stanbic API pilot integrations via their developer portals
- Multi-currency accounts: USD, EUR, GBP accounts tracked alongside UGX
- Automated daily balance snapshot to accounting ledger

Data Integration: Accounting (reconciliation engine)

9

MODULE 6 — CRM & SALES PIPELINE

A full B2B/B2C CRM built for high-velocity sales teams in emerging markets. Supports contact management, pipeline Kanban, deal tracking, and AI-powered lead scoring across multiple country markets.

Key Features

- Contact database: company, person, tags, status badges (Lead/Prospect/Customer/Churned)
- AI Lead Scoring: ARIA scores each lead 0–100 based on engagement, firmographics, and industry data
- Pipeline Kanban: 5-stage drag-and-drop (Prospect → Qualified → Proposal → Negotiation → Won/Lost)
- Deal tracking: value, currency, close date, assigned rep, probability
- Activity feed: calls, emails, meetings, notes — auto-logged from integrations
- Country Analytics tab: heatmap of deals by country/region, revenue concentration
- Email integration (Gmail/Outlook OAuth): bi-directional sync, email templates
- WhatsApp Business API: send templated messages, log conversations to contact record
- Invoice-from-Deal: one-click invoice generation from a won deal
- Custom fields: any contact or deal field can be added without code changes
- Sales forecast: AI-driven weighted pipeline value by period

Data Integration: Accounting (invoice reconciliation), Payments (collection status)

10

MODULE 7 — IT ASSET MANAGEMENT

Enterprise IT governance built for companies managing device fleets, software licences, and network security posture. Critical for growing businesses seeking ISO 27001 readiness.

Key Features

- Asset Register: hardware (laptops, servers, phones, printers), virtual (VMs, cloud resources)
- Asset lifecycle: Procurement → In Use → Maintenance → Disposal with audit trail
- Software licence tracking: licence key, expiry, assigned users, renewal alerts
- Security Health Dashboard: patch compliance %, MFA adoption %, endpoint AV status
- Security health progress bars: colour-coded per department/team
- Vulnerability alerts: CVE feed integration (NVD API), auto-tag affected assets
- Change Management log: ITIL-aligned CAB approvals for infrastructure changes
- IT Support Tickets (integrated helpdesk): submit, assign, track resolution SLA
- Cloud Cost Tracker: AWS/GCP/Azure billing sync — cost-per-project attribution
- Annual IT audit export: generates ISO 27001 evidence pack in PDF

Data Integration: HR (employee ↔ asset assignment), Accounting (asset depreciation)

11

MODULE 8 — CRYPTO EXCHANGE & TREASURY

A dedicated crypto treasury management module enabling businesses to hold, send, receive, and swap digital assets. Built for CFOs who need institutional-grade crypto controls without a separate exchange account.

Key Features

- Multi-chain wallet: BTC (mainnet), ETH (mainnet), USDT (ERC-20, TRC-20), SOL, BNB (BSC)
- Live rate cards: 7-day price history LineChart per asset, powered by CoinGecko API
- Accept Crypto Payment flow: generate payment request → QR code → webhook on confirmation
- Swap (via 1inch aggregator): swap between any two supported assets at best DEX rate
- Crypto payroll option: pay employees in USDT stablecoin (optional, employee-consent required)
- Treasury policy engine: auto-convert crypto receipts to fiat when price crosses threshold
- On-chain analytics: wallet address monitoring, gas fee tracking, nonce management
- Tax reporting: cost-basis tracking (FIFO/LIFO), realised P&L, crypto tax export (CSV)
- AML/KYC: Chainalysis Reactor integration for wallet screening against OFAC/UN sanctions
- Hardware wallet support: Ledger Nano integration for cold-storage treasury balances

Data Integration: Accounting (crypto P&L; journals), Wallet module (unified balance)

12

MODULE 9 — ARIA AI BUSINESS ASSISTANT

ARIA (Advanced Revenue & Intelligence Assistant) is EIC's embedded AI layer. Powered by Anthropic Claude claude-sonnet-4-20250514, ARIA is pre-loaded with the business's live financial context and provides on-demand insights, document generation, anomaly detection, and predictive forecasting.

Key Features

- Conversational interface: full message history, streaming responses, markdown rendering
- System prompt pre-seeded with: current revenue, expenses, tax position, open invoices, cash balance
- Contextual quick-start chips: 'Summarise this month', 'What's my VAT due?', 'Flag anomalies'
- Document generation: draft contracts, reports, letters, emails from natural language prompts
- Anomaly detection: statistical outlier analysis on expense patterns, flagged with explanation
- Cash-flow forecasting: 30/60/90-day rolling forecast with confidence intervals
- Module-aware navigation: ARIA can deep-link user to exact screen based on query
- Multi-language support: English, Swahili, French, Arabic (configurable)
- Voice input (PWA Web Speech API): hands-free queries for field use
- Feedback loop: thumbs up/down per response trains ARIA fine-tuning dataset

Data Integration: All modules (read-only aggregated context injected per session)

13

MODULE 10 — HR & PAYROLL

End-to-end Human Resources management covering employee records, contracts, leave management, and a full payroll engine supporting PAYE, NSSF, and local deductions for Uganda — extendable to any African jurisdiction's payroll rules.

Key Features

- Employee record management: personal details, job history, documents, emergency contacts
- Organisational chart: live drag-and-drop org structure visualisation
- Contract management: offer letters, NDAs, employment contracts — e-signature via DocuSign API
- Leave Management: request, approve, accrual calculations, leave balance dashboard
- Payroll Engine: gross-to-net calculation including PAYE (Uganda tax tables), NSSF (10%), LST, NHIL
- Payslip generation: branded PDF payslips emailed automatically on payroll run
- Bank disbursement: payroll payments via banking module or mobile money (MTN/Airtel bulk disburse)
- PAYE return auto-preparation: monthly URA PAYE return formatted for filing
- Performance Reviews: 360-degree review cycles with scoring rubrics
- Training & Certifications: employee training log, certification expiry alerts
- Workforce analytics: headcount trend, turnover rate, cost-per-employee

Data Integration: Accounting (payroll journals), Banking/Wallet (disbursement), IT (asset assignment)

14

MODULE 11 — INVENTORY & SUPPLY CHAIN

Warehouse and inventory management optimised for African supply chains — handling multi-location stock, import/export tracking, reorder automation, and supplier management with landed-cost calculation.

Key Features

- Product catalogue: SKU, barcode, category, unit of measure, reorder level
- Multi-location stock: warehouse, shop floor, in-transit, consignment tracking
- Purchase Orders: create, approve, receive (3-way match: PO ↔ GRN ↔ Invoice)
- Landed cost engine: adds freight, insurance, customs duty to stock valuation automatically
- Stock movements: adjustments, transfers, write-offs with reason codes and audit trail
- Reorder automation: auto-raise PO when stock falls below reorder point
- Supplier management: supplier scorecard (lead time, defect rate, price history)
- Barcode scanning: web-based scanner via browser camera API
- Inventory valuation: FIFO, LIFO, Weighted Average — switchable per product category
- Low-stock alerts pushed to dashboard and ARIA assistant
- Sales integration: SO fulfilment updates stock in real time

Data Integration: Accounting (stock valuation journals), CRM (sales orders)

15

MODULE 12 — PROJECT MANAGEMENT

A lightweight yet powerful project management module enabling teams to plan, execute, and report on projects — with time tracking, budget monitoring, and resource allocation tied directly to the accounting engine.

Key Features

- Project board: Kanban, List, Gantt (D3-powered) views with drag-and-drop task management
- Task management: assignee, due date, priority, labels, dependencies, sub-tasks
- Milestones: track project phases with burn-down charts
- Time tracking: per-task timers, manual entry, daily timesheets
- Project budget: allocated vs spent vs forecasted — live from accounting integration
- Billable hours: one-click invoice from tracked hours (integrates with Accounting)
- Resource allocation: team capacity planning, utilisation heatmap
- Risk register: log, score (probability × impact), and track mitigation actions
- Status reporting: auto-generate weekly project status report (ARIA-assisted narrative)
- Client portal: read-only project view shareable with external clients via secure link

Data Integration: HR (team members), Accounting (budget vs actuals), CRM (linked to deal)

16

EIC implements a multi-rail payment architecture to maximise acceptance across Africa and globally. No single payment gateway covers all markets, so EIC uses an intelligent routing layer that selects the optimal rail based on sender geography, currency, and payment method — automatically.

Rail	Provider	Coverage	Methods	Settlement
African Mobile Money	Flutterwave	35+ African markets	MTN, Airtel, M-Pesa, Tigo, Orange	1st Centenary Bank
MTN Direct	MTN MoMo API v2	Uganda, Ghana, Cameroon, Ivory Coast	MTN Push/Pull	T+0 sandbox; T+1 prod
Airtel Direct	Airtel Money API	Uganda, Kenya, Zambia, Tanzania, Malawi	Airtel Money	T+1
Card Payments	Stripe	180+ countries	Visa, Mastercard, Amex, Apple Pay	2 Bank, Google Pay
Crypto	Coinbase Commerce - CIPSA	Crypto	BTC, ETH, USDT, SOL, BNB, LTC	Blockchain; fiat T+1
Bank Transfer	Pesalink (Kenya) / RTGS (Uganda)	East (Uganda) SWIFT global	Bank-to-bank EFT/RTGS	T+0 same-bank; T+1 cross-bank
BNPL / Credit	Lipa Later (Africa)	Kenya, Uganda, Tanzania	Buy Now Pay Later	T+0 to merchant

- 1. Customer Initiates:** Customer selects payment method on EIC checkout / invoice payment page.
- 2. Gateway Selection:** EIC Payment Router evaluates geography, currency, and preferred method to select optimal gateway.
- 3. Payment Request:** EIC backend calls selected gateway API (Flutterwave/Stripe/MTN SDK) with encrypted payload over TLS 1.3.
- 4. Customer Authenticates:** MTN/Airtel: USSD PIN. Stripe: 3D Secure. Crypto: wallet signature. All 2FA where supported.
- 5. Webhook Received:** Gateway POSTs event to EIC webhook endpoint (/api/payments/webhook) over HTTPS. Signature verified using HMAC-SHA256.
- 6. Idempotency Check:** EIC checks idempotency key against Redis cache — duplicate webhooks are discarded.
- 7. Ledger Posting:** Accounting Engine posts journal: Dr Accounts Receivable / Cr Revenue + Cr VAT Payable.
- 8. EFRIS e-Receipt:** URA EFRIS API called to issue e-receipt. Receipt number and QR returned and stored.
- 9. Settlement:** Funds settle to Centenary Bank Uganda account (****8770) via Flutterwave settlement cycle.
- 10. Notification:** Customer receives email + SMS + WhatsApp confirmation with receipt. EIC dashboard updates in real time via WebSocket.

All credentials are stored exclusively in server-side environment variables (never in frontend source). Production secrets are managed via HashiCorp Vault. The following keys are required:

[illegible]

[illegible]

17 PCI DSS COMPLIANCE FRAMEWORK

PCI DSS v4.0 Strategy

EIC never stores, processes, or transmits raw cardholder data. By using Stripe's hosted payment fields (Stripe.js) and Flutterwave's PCI-certified hosted checkout, EIC qualifies for SAQ-A (the lightest PCI questionnaire), reducing compliance burden by over 90%. The following table maps EIC's architecture to PCI DSS v4.0 requirements.

PCI Req.	Requirement Description	EIC Implementation	Status
1	Install/maintain network security controls	Cloudflare WAF + Kubernetes network policies + private network	PASS/PC
2	Apply secure config to all system components	OpenSCAP benchmark hardening on all container images; CIS benchmarks	PASS
3	Protect stored account data	No PANs stored. Stripe/Flutterwave tokenisation only. Tokens AES-256 encrypted at rest	PASS
4	Protect data in transit	TLS 1.3 enforced on all endpoints. HSTS headers. Certificate pinning in mobile app	PASS
5	Protect against malware	Container image scanning (Trivy). Dependency vulnerability scanning (Snyk). Runtime RASP	PASS
6	Develop/maintain secure systems	OWASP Top 10 code review checklist. SAST (CodeQL), DAST (OWASP ZAP in CI pipeline)	PASS
7	Restrict access by business need to BAOs	RBAC with principle of least privilege. All prod access via HashiCorp Vault + SSO with MFA	PASS
8	Identify users and authenticate access	JWT + refresh tokens. MFA mandatory for admin roles. Biometric on mobile	PASS
9	Restrict physical access	Hosted on AWS/Cloudflare — SOC 2 Type II certified. Data centers	DEFERRED
10	Log and monitor all access to systems	Datadog APMs + CloudTrail. Immutable audit logs. 1 year retention	PASS
11	Test security regularly	Quarterly pen-tests. Annual PCI ASV scan. Continuous monitoring in staging	PASS
12	Support information security with policies	Security Policy, Incident Response Plan, Data Retention Policy, etc. Documented	PASS

Tokenisation & Card Data Isolation

EIC uses Stripe Elements and Flutterwave inline checkout iframes for all card entry. These are cross-origin iframes served directly from the payment processor's PCI-certified domain — EIC's JavaScript never has access to raw card numbers. Stripe returns a payment_method token; EIC stores only this token (not a card number) for recurring billing. All tokens are encrypted with AES-256-GCM before database persistence.

SECURITY: Bank account 32046608770 is stored ONLY in server-side .env / HashiCorp Vault. All UI surfaces display masked value ****8770. No

18 SECURITY ARCHITECTURE

Defence-in-Depth Strategy

EIC applies a layered security model with six distinct defence rings. A breach at any one layer does not compromise the system — each layer is independently hardened and monitored.

Ring	Layer	Controls Implemented
1 — Perimeter	Network & CDN	Cloudflare WAF rules, DDoS mitigation (Layer 3/7), geo-blocking, bot scoring, rate limiting (100 req/min p
2 — Transport	TLS Everywhere	TLS 1.3 only, HSTS with preload, OCSP stapling, certificate auto-renewal (Let's Encrypt / Cloudflare)
3 — Application	Input Validation	Zod schema validation on all API inputs. Parameterised SQL (Drizzle ORM — zero raw SQL). XSS CSP P
4 — Authentication	Identity	JWT (15-min expiry) + Refresh tokens (7-day, rotated on use). OAuth2 for social login. TOTP/FIDO2 MFA
5 — Authorisation	Access Control	RBAC: Owner, Admin, Manager, Accountant, Viewer. PostgreSQL Row-Level Security enforces tenant is
6 — Data	Encryption & Privacy	AES-256-GCM for sensitive fields (bank accounts, API keys). bcrypt (rounds=12) for passwords. GDPR-c

OWASP Top 10 Mitigations

- A01 Broken Access Control:** PostgreSQL RLS + RBAC + route guards. All API endpoints declare required roles. Automated tests assert 401/403 on privilege escalation attempts.
- A02 Cryptographic Failures:** TLS 1.3. AES-256-GCM field encryption. No MD5/SHA-1. JWT uses RS256 (asymmetric signing). Password hashing with bcrypt rounds=12.
- A03 Injection:** Drizzle ORM parameterised queries. Zod input sanitisation. Content-Security-Policy blocks inline scripts. No eval() anywhere in codebase.
- A04 Insecure Design:** Threat modelling per module during design. Minimal data collection (data minimisation). Separate staging environment mirrors production.
- A05 Security Misconfiguration:** Infrastructure-as-code (Terraform). All defaults changed. Security headers (Helmet.js). Docker images rebuilt weekly with latest base.
- A06 Vulnerable Components:** Snyk SCA in CI/CD. Dependabot PRs auto-opened for CVE-scored deps. Container scanning (Trivy) before every deployment.
- A07 Auth & Session Failures:** JWT short-lived (15 min). Refresh token rotation + revocation list in Redis. Concurrent session limits. Brute-force logout after 5 attempts.
- A08 Software & Data Integrity:** npm lockfile committed. Subresource Integrity on CDN assets. Signed Docker images. Code-signing on deployment artifacts.
- A09 Security Logging Failures:** Every auth event, permission change, payment attempt logged to Datadog. Immutable audit trail in PostgreSQL. Log integrity via hash chaining.
- A10 Server-Side Request Forgery:** Allowlist of permitted external URLs. No user-supplied URLs passed to backend HTTP clients. Kubernetes network policies block metadata endpoint.

19 DATABASE SCHEMA & API DESIGN

Core Database Tables

EIC uses a single PostgreSQL 16 database with Row-Level Security (RLS) policies for multi-tenancy. The schema is managed via Drizzle ORM migrations. Below are the primary entity groups.

Table	Key Columns	Relationships & Notes
organisations	id, name, plan_tier, country, currency, created_at	Root tenant entity. All other tables FK to org_id
users	id, org_id, email, password_hash, role, mfa_secret	RBAC roles: owner admin manager accountant viewer
accounts	id, org_id, code, name, type, parent_id, currency	Chart of accounts; type: asset liability equity revenue expense
journal_entries	id, org_id, date, reference, description, status, posted_by	Double-entry header; status: draft posted reversed
journal_lines	id, entry_id, account_id, debit, credit, currency, fx_rate	Minimum 2 lines per entry; debits = credits enforced by trigger
invoices	id, org_id, contact_id, number, date, due_date, status, total	status: draft sent paid overdue voided
contacts	id, org_id, type, name, email, phone, country, type	types: customer supplier both
transactions	id, org_id, type, provider, ref, amount, currency, attached_events	Payments & events; provider: stripe flutterwave mtn airtel crypto
crypto_wallets	id, org_id, chain, address, balance, last_sync	One wallet per chain per org
inventory_items	id, org_id, sku, name, uom, cost_price, sell_price, stock_level	Stock levels updated via movement triggers
employees	id, org_id, user_id, department, job_title, salary, linked_to, status	linked_to: user_id for system access
payroll_runs	id, org_id, period_start, period_end, status, total_gross, draft	status: draft approved paid reversed
projects	id, org_id, name, client_id, budget, currency, status, start_date, end_date	
tasks	id, project_id, title, assignee_id, status, priority, due_date, hours_logged	
assets_it	id, org_id, name, type, serial, assigned_to, purchase_price	IT Asset Register
efris_submissions	id, org_id, type, period, status, ura_ref, submitted_type, auto_return	types: return invoice receipt
audit_logs	id, org_id, user_id, action, entity, entity_id, old_value, new_value	Immutable, INSERT only; no UPDATE/DELETE

REST API Design Principles

- Base URL: https://api.eic.global/v1/
- Authentication: Bearer JWT in Authorization header on all /v1/* routes
- Versioning: URI versioning (/v1/, /v2/) — never break existing clients
- Pagination: cursor-based (after=&limit;=50) for all list endpoints
- Filtering: query params follow JSON:API spec (?filter[status]=active)
- Errors: RFC 7807 Problem Details (application/problem+json) with error code, message, path
- Rate Limits: 1,000 req/min per API key (Enterprise: 10,000). Headers: X-RateLimit-Remaining
- Idempotency: POST endpoints accept Idempotency-Key header to prevent duplicate processing
- Webhooks: all async events POSTed to customer-registered URL with HMAC-SHA256 signature
- OpenAPI 3.1 spec auto-generated from Zod schemas via zod-to-openapi. Published at /v1/openapi.json

20 DEVOPS, CI/CD & INFRASTRUCTURE

Infrastructure as Code

Every infrastructure resource is defined in Terraform. No manual console changes permitted in production. State stored in Terraform Cloud with mandatory plan-before-apply workflow requiring two approvals for production changes.

Component	Tool / Provider	Configuration
Container Orchestration	Kubernetes (GKE Autopilot)	3-zone HA; auto-scaling 2–20 pods per service; pod disruption budgets
Container Registry	Google Artifact Registry	Private; image signing; vulnerability scanning on push
CI/CD Pipeline	GitHub Actions	lint → test → build → SAST → deploy-staging → smoke-test → deploy-prod
Secret Management	HashiCorp Vault	Dynamic DB credentials; PKI; all secrets injected at runtime; no env files on disk
Database	Cloud SQL (PostgreSQL 16)	Multi-zone HA; automated backups every 6 hours; point-in-time recovery 30 days
Cache	Cloud Memorystore (Redis 7)	HA cluster mode; auth enabled; TLS in transit
CDN / Edge	Cloudflare Enterprise	Global PoP; Smart Routing; R2 object storage; WAF; Zero Trust Network Access
DNS	Cloudflare DNS	DNSSEC enabled; anycast routing; 100% uptime SLA
Monitoring	Datadog APM + Logs + Synthetics	Custom dashboards per module; SLO tracking; PagerDuty escalation
Error Tracking	Sentry	Source maps uploaded; performance monitoring; release tracking
Load Testing	k6 + Grafana Cloud	Pre-release load tests: 10,000 concurrent users target
Backup	GCS + AWS S3 cross-region	Daily DB dumps encrypted with customer-managed KMS keys; 90-day retention

CI/CD Pipeline Detail

- Code Push:** Developer pushes feature branch → GitHub PR opened
- Lint + Type Check:** ESLint (frontend), tsc --noEmit, Biome formatter — fail fast
- Unit + Integration Tests:** Jest (backend) + Vitest (frontend) — minimum 80% coverage gate
- SAST Scan:** CodeQL + Semgrep — blocks merge on HIGH/CRITICAL severity findings
- Docker Build:** Multi-stage Dockerfile: build → prune devDeps → distroless runtime image
- Container Scan:** Trivy scans image for CVEs — blocks on CRITICAL vulnerabilities
- Deploy to Staging:** Helm chart upgrade on staging K8s namespace — zero-downtime rolling deploy
- Smoke Tests:** Playwright E2E suite: auth flow, payment flow, invoice creation — 15 critical paths
- Performance Gate:** k6 quick load test (500 VUs, 60s) — p99 latency < 800ms gate
- Deploy to Production:** Argo CD GitOps sync — blue/green deployment; instant rollback on health fail
- Post-Deploy Verification:** Datadog Synthetics runs full suite against production — PagerDuty alert on fail

21 GO-TO-MARKET & PRICING STRATEGY

Market Entry Strategy

EIC will enter the market with a Uganda-first strategy — leveraging the founder's local network, EFRIS compliance as a regulatory wedge, and mobile money as the default payment rail. Phase 1 focuses on acquiring 1,000 paying Ugandan SMEs within 12 months. Phase 2 expands to Kenya, Nigeria, and Ghana. Phase 3 targets global expansion via partnerships.

Pricing Tiers (Detailed)

Feature	Starter \$29	Growth \$99	Business \$299	Enterprise Custom
Users	5	25	100	Unlimited
Dashboard	Yes	Yes	Yes	Yes
Accounting (Full)	Yes	Yes	Yes	Yes
EFRIS / URA Tax	Yes	Yes	Yes	Yes + multi-country
Mobile Money	Yes	Yes	Yes	Yes
Banking Integration	Limited (1 account)	Yes (3 accounts)	Yes (unlimited)	Yes + API
CRM	Basic	Full + AI scoring	Full + Analytics	White-label
HR & Payroll	No	Yes (25 employees)	Yes (100 empl.)	Unlimited
Inventory	No	Yes	Yes	Yes + EDI
IT Asset Mgmt	No	No	Yes	Yes
Crypto Wallet	No	Yes (view only)	Yes (full)	Yes + treasury
Project Management	No	No	Yes	Yes + portfolio
ARIA AI	10 queries/day	100 queries/day	Unlimited	Custom fine-tune
API Access	No	Read-only	Full	Full + webhooks
Data Residency	Shared	Shared	Dedicated schema	Dedicated instance
SLA	99.5%	99.9%	99.95%	99.99% + credits
Support	Community + email	Email (24h SLA)	Priority (4h SLA)	Dedicated CSM
Onboarding	Self-serve	Guided setup	Full onboarding	White-glove

Revenue Projections (Conservative)

Period	Starter Customers	Growth Customers	Business Customers	MRR (USD)	ARR (USD)
Month 3	50	10	2	\$2,848	\$34,176
Month 6	150	40	10	\$10,290	\$123,480
Month 12	400	120	35	\$28,765	\$345,180
Year 2	1,200	400	120	\$94,800	\$1,137,600

Period	Starter Customers	Growth Customers	Business Customers	MRR (USD)	ARR (USD)
Year 3	3,500	1,200	400	\$282,100	\$3,385,200

Distribution Channels

Direct (Web): SEO-optimised marketing site at www.eic.global. Google Ads targeting 'accounting software Uganda/Africa'. Free trial CTA.

Partnerships — Telcos: Bundled trial with MTN MoMo Business and Airtel Business subscriptions. Co-marketing budget.

Partnerships — Banks: White-label for Centenary Bank and DFCU Bank SME banking clients. Revenue share model.

Reseller Programme: Certified accountants and IT consultants earn 25% recurring commission on referred clients.

App Marketplaces: Listing on Microsoft AppSource, Shopify App Store (for inventory module), and Zoho Marketplace.

Content Marketing: Engola Innocent thought-leadership on LinkedIn targeting African CFOs. YouTube tutorials in English + Swahili.

Events: Uganda National Chamber of Commerce sponsorship. Demo booths at Kampala TechForum and Africa Fintech Summit.

22

BUILD PHASES & TIMELINE

12-Month MVP Roadmap

The build is structured in 4 phases. Each phase ends with a shippable product increment that can be validated with real customers before the next phase begins. This de-risks the build and allows early revenue generation.

Phase 1 — Foundation & Core Finance (Weeks 1–10)

- Project scaffolding: monorepo (Turborepo), Docker Compose dev environment, CI/CD pipeline
- Authentication system: registration, login, MFA, OAuth2 (Google), JWT + refresh token rotation
- Tenant management: organisations, user roles, settings, billing portal (Stripe billing)
- Smart Dashboard: layout, sidebar, header, StatCards, basic charts with mock data
- Accounting & Finance: Chart of Accounts, double-entry journal engine, invoicing, basic reporting
- EFRIS Integration: e-invoice, e-receipt, VAT return prep, direct URA API submission
- Postgres schema + Drizzle migrations for all Phase 1 entities
- Unit tests (80%+ coverage), integration tests for all API routes
- Staging environment on GKE with domain eic-staging.global

Phase 2 — Payments & Mobile Money (Weeks 11–20)

- Payment Gateway integration: Flutterwave (MTN MoMo, Airtel, cards), Stripe (global cards)
- MTN MoMo direct API: collections, disbursements, balance, transaction status
- Airtel Money direct API: same capability set
- Webhook handler: signature verification, idempotency, retry queue
- Banking Integration module: account cards, statement import, bank reconciliation
- Crypto Wallet module: Coinbase Commerce, live price feeds (CoinGecko), wallet display
- Settlement automation: Flutterwave → Centenary Bank reconciliation
- PCI SAQ-A attestation: Stripe Elements + Flutterwave inline checkout implementation
- End-to-end payment tests with sandbox credentials for all providers

Phase 3 — Operations Modules (Weeks 21–34)

- CRM: contacts, pipeline Kanban, lead scoring, activity feed, WhatsApp Business API integration
- HR & Payroll: employee records, leave management, payroll engine (PAYE/NSSF Uganda), payslips
- Inventory & Supply Chain: product catalogue, stock management, purchase orders, 3-way match
- IT Asset Management: asset register, licence tracking, security health dashboard
- Project Management: Kanban/Gantt board, task management, time tracking, budget monitoring
- ARIA AI Assistant: Anthropic API integration, system prompt engineering, streaming responses
- Multi-currency: FX rate integration, currency-aware charts and reports
- Mobile app (React Native): iOS + Android build, mobile-optimised dashboard + payments

Phase 4 — Scale, Polish & Launch (Weeks 35–48)

- Crypto Exchange module: swap (1inch), treasury policy engine, AML screening (Chainalysis)
- Advanced reporting: custom report builder, scheduled email reports, executive PDF packs
- Multi-country tax engine: Kenya (KRA), Nigeria (FIRS) adapters
- Performance optimisation: sub-2s TTI, 90+ Lighthouse score, PWA offline mode
- Security audit: third-party penetration test, OWASP ASVS Level 2 assessment

- Load testing: 10,000 concurrent users target with k6; infrastructure auto-scaling verified
- WCAG 2.1 AA accessibility audit and remediation
- Marketing site (www.eic.global): hero, features, pricing, demo booking, blog CMS
- Public API + developer portal (developers.eic.global): OpenAPI docs, SDK (Node.js + Python)
- Go-live: DNS cutover, monitoring dashboards live, on-call rotation established
- Post-launch: weekly sprints based on user feedback; 99.9% uptime target enforced

Team Composition (Recommended)

Role	Count	Responsibility	Phase
Full-Stack Lead (TypeScript)	1	Architecture, code review, core modules	All
Frontend Engineer	2	React UI, component library, charts, animations	1–4
Backend Engineer	2	APIs, payment integrations, database, auth	1–4
DevOps / Platform Engineer	1	K8s, CI/CD, monitoring, security hardening	1–4
Mobile Engineer (React Native)	1	iOS + Android app	3–4
QA Engineer	1	Test automation, regression suites, load testing	2–4
UI/UX Designer	1	Design system, user flows, prototyping	1–2
Product Manager	1	Roadmap, user research, backlog management	All

23

PRE-LAUNCH CHECKLIST

Production Readiness Gate

Every item below must be completed and signed off by a responsible owner before production launch. No item is optional. Partial completion does not constitute readiness.

Security & Compliance

- All .env secrets migrated to HashiCorp Vault — no secrets in codebase
- Third-party penetration test completed — all HIGH/CRITICAL findings remediated
- PCI SAQ-A attestation signed and filed
- GDPR Data Processing Agreement in place with all sub-processors
- Privacy Policy and Terms of Service reviewed by licensed legal counsel
- Centenary Bank account confirmed and settlement tested end-to-end
- AML/KYC policy documented and screening tool (Chainalysis) active

Infrastructure & Performance

- HTTPS + TLS 1.3 enforced on all public endpoints — HTTP redirects to HTTPS
- HSTS header with preload submitted to browser preload list
- Load test completed: 10,000 concurrent users — p99 < 1,000ms
- Auto-scaling verified: pods scale up and down correctly under load
- Database backups tested: point-in-time restore completed successfully
- Disaster recovery runbook documented and rehearsed
- CDN and WAF rules tuned and validated
- Monitoring dashboards live: Datadog SLO, Sentry error budget, PagerDuty on-call active

Payment & Finance

- MTN MoMo production credentials tested with live transactions
- Airtel Money production credentials tested end-to-end
- Flutterwave production webhook verified and idempotency confirmed
- Stripe production keys active — 3DS2 flows tested on multiple card networks
- Coinbase Commerce webhooks verified — crypto payment confirmed on testnet then mainnet
- URA EFRIS production TIN and Device Number validated — e-invoice issued successfully
- Bank reconciliation: first live statement import and auto-match verified

Quality & Accessibility

- Unit test coverage $\geq 80\%$ (backend) and $\geq 75\%$ (frontend)
- Playwright E2E suite: all critical user journeys passing
- Cross-browser testing: Chrome, Firefox, Safari, Edge, Opera Mini (low-bandwidth)
- Mobile responsiveness verified: 375px, 768px, 1280px, 1920px viewports
- WCAG 2.1 AA audit passed — all Level A + AA criteria met
- Screen reader testing completed (NVDA + VoiceOver)
- CEO and team profile page reviewed and approved by Engola Innocent

Business & Operations

- Marketing site live at www.eic.global with pricing, features, and demo booking
- Support email (support@eic.global) configured and routed to helpdesk
- On-boarding flow tested end-to-end: registration → first invoice → first payment
- Billing system verified: Stripe subscription billing cycles tested for all tiers
- API documentation published at developers.eic.global
- Status page live at status.eic.global (Statuspage.io)

24 LEGAL, PRIVACY & REGULATORY

Legal & Regulatory Framework

EIC operates at the intersection of financial software, payments, and AI. This requires careful navigation of multiple regulatory frameworks across all operating jurisdictions.

Regulation / Framework	Jurisdiction	EIC Obligation	Status
Uganda Data Protection Act 2019	Uganda	Register as data controller with PDPO. Privacy policy and data subject rights portal.	Required
GDPR (EU General Data Protection Regulation)	EU / Global	DPA with EU sub-processors. GDPR-compliant contracts and data residency options for EEA customers.	Required
PCI DSS v4.0 (SAQ-A)	Global	Annual SAQ-A self-assessment. Quarterly ASV scans. No raw payment data stored.	Required
URA EFRIS Compliance	Uganda	Licensed EFRIS integrator. Device registration. Record-keeping and submission.	Required
Bank of Uganda EMI Regulations	Uganda	For mobile money aggregation: EMI licence or partnership with licensed entity (Flutterwave).	Required
FCA Electronic Money Regs 2011	UK	Required if accepting GBP payments or serving UK customers. Stripe handles as licensed.	Required
FinCEN / BSA (USA)	USA	US AML programme if serving US customers. SAR filing obligations. Consider MSB registration.	Required
VASP Regulations	Uganda / Global	Crypto payments require Virtual Asset Service Provider registration in applicable jurisdiction.	Required
IFRS Financial Reporting	Global	Accounting engine outputs must align with IFRS standards for professional audit readiness.	Required
WCAG 2.1 AA	Global	Accessibility compliance reduces legal exposure in EU, US, and UK markets.	Required

Intellectual Property

The EIC platform, including all source code, design assets, brand marks, trade names, and documentation, is the exclusive intellectual property of Engola Innocent / EIC Enterprise Ltd. All employees and contractors must sign a Proprietary Information and Invention Assignment (PIIA) agreement before commencing work. The EIC name and logo must be registered as trademarks in Uganda (URSB), Kenya (KEIPO), and internationally (WIPO Madrid Protocol) before public launch.

Final Vision Statement

"EIC is not just software. It is the financial nervous system of every business on earth. Built by Engola Innocent with a singular mission: to give every entrepreneur, every SME, every corporation — regardless of their country, language, or currency — the same world-class tools that were previously only available to Fortune 500 companies. One platform. Every business. Every country. Every currency. Forever."

www.eic.global | support@eic.global | developers.eic.global

Built by EIC. Owned by the world.